

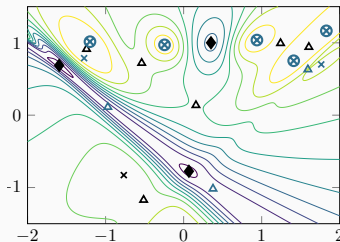
Specific Euclidean Distance Degrees of Secant Varieties to the Rational Normal Curve

From Geometry to Numbers: a celebration of women in mathematics

Leonie Kayser & Sibren Lagauw

leokayser.github.io

May 27, 2026



MAX PLANCK INSTITUTE
FOR MATHEMATICS
IN THE SCIENCES

Euclidean Distance Degree

- ▷ Given a variety $X \subseteq \mathbb{C}^N$ and a point $y \in \mathbb{R}^N$, find closest point on $X(\mathbb{R})$
- ▷ Distance measured using non-degenerate (real) quadric $Q(x) = x^\top \Lambda x$

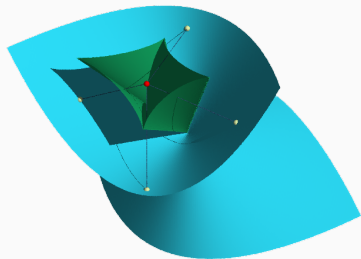
Definition (Euclidean distance degree, $\text{EDD}_Q(X)$)

The number of complex critical points of $\hat{y} \mapsto Q(\hat{y} - y)$ on X_{reg} for general $y \in \mathbb{R}^N$ is the **Euclidean Distance degree** of X . $\text{EDD}_Q(X \subseteq \mathbb{P}^{N-1}) := \text{EDD}_Q(\hat{X})$.

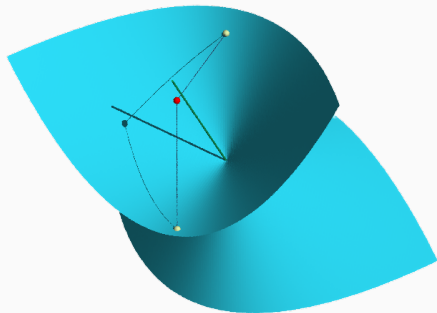
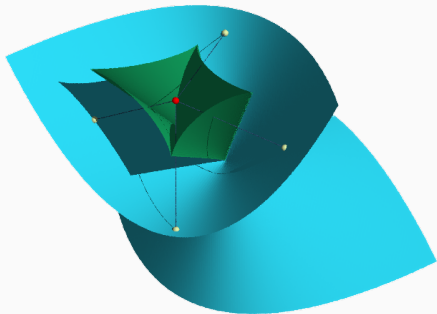
- ▷ Formally: Consider **distance correspondence**

$$\mathcal{E}_X = \overline{\{(y, \hat{y}) \mid \hat{y} \in X_{\text{reg}}, y - \hat{y} \perp_Q T_{\hat{y}}X\}} \subseteq \mathbb{C}^N \times X$$

- ▷ Projection $\pi_1: \mathcal{E}_X \rightarrow \mathbb{C}^N$ is generically finite,
 $\text{EDD}_Q(X) := \deg \pi_1$ (or 0 if π_1 not dominant)
- ▷ If $Q \succ 0$ and $X_{\text{reg}}(\mathbb{R}) \neq \emptyset$, then $\text{EDD}_Q(X) > 0$



Generic vs specific EDD



- ▷ For general quadric obtain **generic EDD**, $\text{EDD}_{\text{gen}}(X) \geq \text{EDD}_Q(X)$ for all Q
- ▷ EDD_{gen} of $X \subseteq \mathbb{P}^d$ is determined by Chern(-Mather) classes

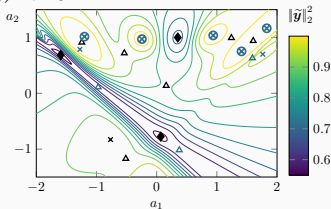
$$\text{EDD}_{\text{gen}}(X) = \sum_{i=0}^{\dim X} (-1)^i (2^{\dim X + 1 - i} - 1) \deg_{\mathbb{P}^d} c_i^{\text{M}}(X).$$

Secant varieties of the rational normal curve

- ▷ Rational normal curve $X_{d,1} = \nu_d \mathbb{P}^1 = \text{Im}(x \mapsto (x_0^d : x_0^{d-1}x_1 : \cdots : x_1^d)) \subseteq \mathbb{P}^d$
- ▷ Secant variety $\sigma_r X = \overline{\bigcup_{p_1, \dots, p_r \in X} \langle p_1, \dots, p_r \rangle} \subseteq \mathbb{P}^d$, together

$$X_{d,r} := \sigma_r \nu_d \mathbb{P}^1 = \left\{ y \in \mathbb{P}^d \mid \text{rank} \begin{bmatrix} y_0 & y_1 & \cdots & y_r \\ y_1 & y_2 & \cdots & y_{r+1} \\ \vdots & \vdots & \ddots & \vdots \\ y_{d-r} & \hat{y}_{N-r} & \cdots & y_d \end{bmatrix} < r \right\} \subseteq \mathbb{P}^d$$

- ▷ $X_{d,r} \subseteq \mathbb{P} \text{Sym}^d \mathbb{C}^2$ is the variety of binary forms of (border)rank at most r
- ▷ Distance optimization to $X_{d,r}(\mathbb{R})$ relevant in systems engineering
- ▷ Interesting quadrics: Unit $\mathbf{1}_{d+1}$, Bombieri $\Theta := \text{diag}(\binom{d}{i})_{i=0}^d, \dots$
- ▷ Previous work on $\text{EDD}_Q(X_{d,r})$
 - [Ottaviani-Spaenlehauer-Sturmfels 2014] generic Q
 - [Panah 2025] Bombieri norm Θ , $r = d/2$
 - [Kozhasov-Muniz-Qi-Sodomaco 2025] all Q , $r = 1$



Main results

Theorem ([Kayser–Lagauw 2026+])

1. Give simple polynomial system whose roots correspond to critical points, “optimally” solvable using parameter continuation algorithms.
2. Explicitly construct determinantal scheme $\mathcal{B}_\Lambda \subseteq \mathbb{P}^r$ such that if $\mathcal{B}_\Lambda = \emptyset$, then $\text{EDD}_Q(X_{d,r}) = \text{EDD}_{\text{gen}}(X_{d,r})$.
3. Obtain new expression via Thom–Porteous formula

$$\text{EDD}_{\text{gen}}(X_{d,r}) = \sum_{j=1}^r \binom{d-2r+j}{j} 3^j = \sum_{i=1}^r \binom{d-r+1}{r-i} \binom{d-2r+i}{i} 2^i.$$

4. If $\mathcal{B}_\Lambda$ is non-empty but finite, then $\deg \mathcal{B}_\Lambda \leq \text{EDD}_{\text{gen}}(X_{d,r}) - \text{EDD}_Q(X_{d,r})$, can computationally certify if equality holds.

EDD_Q(X_{d,r}) for special quadrics

	$\Lambda = 1$ Unit				$\Lambda = F$ Frobenius				$\Lambda = \Theta$ Bombieri			
$d \setminus r$	1	2	3	4	1	2	3	4	1	2	3	4
1	1				1				1			
2	4				2				2			
3	7	1			7	1			3	1		
4	10	13			6	9			4	7		
5	13	34	1		13	34	1		5	16	1	
6	16	64	40		10	38	34		6	28	20	
7	19	103	142	1	19	103	142	1	7	<u>43</u> .. <u>45</u>	<u>62</u> .. <u>64</u>	1
8	22	151	334	121	14	103	246	113	8	<u>61</u> .. <u>65</u>	<u>134</u> .. <u>142</u>	53
9	25	208	643	547	25	208	643	543	9	<u>82</u> .. <u>88</u>	<u>243</u> .. <u>263</u>	229

► Method recovers *all* previously known values for EDD_Θ(X_{d,r}), additionally

$$\text{EDD}_{\Theta}(X_{5,2}) = 16, \quad \text{EDD}_{\Theta}(X_{6,2}) = 28$$

Thank you! Questions?

Image credit

- ▶ Slide 1,2: Thanks to Luca Sodomaco for letting me use his graphics!
- ▶ Slide 0,3: Contour plot by Sibren Lagauw using MatLab and pgfplots